

AI & Technology Landscape — September 2026

The September frontier wave, and what is now buyable or buildable for CNC from DXF, drawing packs from SOLIDWORKS, estimating, and the in-house PDM

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Owner: James Gray

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Twelve days after the August issue, the first week of September delivered the densest frontier release window of the year: **Claude Fable 5.1** (1 Sep), **Gemini 3.8 Flash** (2 Sep), **Meta Muse Spark 1.3** (2 Sep) and **GPT-6 Astra** (3 Sep). None of them changes the engine's architecture, and this report says so plainly. What it adds is four sections James asked for by name: **G-code from DXF for the CNC workstream**, **drawing packs from SOLIDWORKS**, **the estimating market**, and **replicating PDM in-house** — each with a verdict and a next step rather than a list.

Three findings are worth reading before the rest. **The shared workstation ordered this week carries a 16 GB card, and the on-premise vision model this report recommended in August needs about 17 GB** — decide that before the order goes. **"AI G-code" is a milling story in 2026**: every AI CAM product on the market is 3-axis or 3+2 milling, and the honest route for a TRUMPF laser is DXF into TruTops Boost's own automation, which SDI already owns. And **Werk24 reads a PDF drawing into structured JSON for under a pound a page**, which is the first product close enough to the engine's own readers to be worth an afternoon as a comparator.

On sourcing. **VENDOR** entries come from the vendor's own pages. **SECONDARY** entries come from third-party write-ups and want confirming against vendor pricing before anything is committed. **On the August actions:** the code says the two model-string swaps have not been made — the engine still calls **grok-4.3** for vision and **claude-sonnet-4** on the Anthropic path — and no report back is recorded on the LEO, Anchorpoint or CADshift items. They are carried below, not repeated as new.

THE FORTNIGHT IN ONE SCREEN

What shipped, and whether it changes anything for us

DATE	WHAT	WHO	WHY IT MATTERS HERE	VERDICT
1 Sep	Claude Fable 5.1 / Mythos 5.1	Anthropic	Same \$10/\$50 per 1M as Fable 5; cache reads cut 75% to \$0.25 ; 1M context; three breaking API changes (forced	NO CHANGE

DATE	WHAT	WHO	WHY IT MATTERS HERE	VERDICT
			tool choice now errors; thinking blocks not readable by earlier models; editing earlier turns invalidates later thinking). The engine's Anthropic call is a plain message with no forced tool use, so none of the three bite — but the engine's target on this path is Opus 5, not Fable.	
2 Sep	Gemini 3.8 Flash	Google	Same introductory \$0.75/\$3.75 to 31 Dec, then \$1.50/\$7.50. Native PDF input alongside image, 1M context, 64K output, thinking level low/medium/high. Batch and Flex at half price (\$0.375/\$1.875). Reported to beat Opus 5 on three of Google's published benchmarks.	TRIAL · MULTI-PDF
3 Sep	GPT-6 Astra	OpenAI	\$10/\$50 per 1M, cached \$1, batch/flex half, 1.05M context, image input, effort low→max. Preview to partners 3 Sep, paid users 4 Sep, with cyber	NO CHANGE

DATE	WHAT	WHO	WHY IT MATTERS HERE	VERDICT
			capabilities gated. Frontier price for a path where the engine runs <code>gpt-4.1-mini</code> as a price-lookup fallback. Nothing to move for.	
2 Sep	Muse Spark 1.3	Meta	Multimodal reasoning model, 1M context, in Muse Code and the Meta Model API. Not open-weights ; Meta says an open release is "coming soon" with no date or licence. A Contributor tier trains on your data; the Standard tier does not. Watch for the weights, nothing else.	WATCH
1 Sep	PPLX Qwen 3.8 27B, local	Perplexity	Perplexity's Hybrid Compute runs a post-trained Qwen3.8-27B <i>on the user's own machine</i> for the private half of a task, gated by an on-device PII classifier. Evidence that the 27B local class is production-grade, from a company with no reason to flatter it.	SUPPORTS AUG CALL
Oct	SOLIDWORKS 2027	Dassault	Preview ran mid-July to mid-	PLAN FOR IT

DATE	WHAT	WHO	WHY IT MATTERS HERE	VERDICT
			August; SOLIDWORKS Live event 15 Oct. LEO generates a drawing from a model by conversation — template, standard, primary view, requirements — and saves it. Sheet-metal stamp-on-bend. A new Document Manager API key is needed per major release, which touches P1 and P2.	
Now	Werk24	W24 Service GmbH	API reads a PDF or image drawing and returns JSON: title block (ID, material, weight, tolerances, BOM, notes), dimensions with confidence and bounding boxes, GD&T, threads, processes to DIN 8580, bounding-box and feasibility signals. From €0.95 a page, no commitment; 5-20 s a page.	LOOK AT THIS
Carried	Opus 5 and Grok 4.6 swaps	Us	Recommended 26 Aug; not made. <code>config.py</code> and <code>llm_scan_price.py</code> still default to <code>grok-4.3</code> ;	DO ON A REPLAY

DATE	WHAT	WHO	WHY IT MATTERS HERE	VERDICT
			web_ai_price_lookup.py still names claude-sonnet-4-20250514 . Do them on a replay, not on live parallel-run jobs — see the frontier section.	

ON-PREMISE VISION, REVISITED — THE CARD IN THE ORDER IS 16 GB

DECIDE THIS WEEK, BEFORE THE WORKSTATION ORDER GOES

The August recommendation and the September order do not fit each other

August said: Qwen3.8-27B, 4-bit, fits a 24 GB card; price a card and benchmark it on 10575-02. The shared workstation specification of 3 September, going to order this week, carries an **NVIDIA RTX PRO 2000 or RTX 2000 Ada, 16 GB** — chosen for Drawing Search's 9 GB resident index and because it is SOLIDWORKS-certified. Both reasons are good. But secondary write-ups put the 4-bit 27B at **about 17 GB**, and that is before the vision encoder's working memory and before Drawing Search's index shares the card.

So the local-vision trial does not run on this box as specified. That is not an argument against the box; it is an argument for deciding on purpose. Three honest options, in the order I would take them:

1. Order as specified, and keep vision hosted. Vision at rank 40 costs pence a page today and the engine's determinism proof does not depend on it. Revisit when a card is needed for something else. **2. Step to a 24 GB certified card** (RTX PRO 4000 class) and run the trial on the same machine — a few hundred pounds on a £4.8–5.05k request, and the PSU headroom the spec already allows for. **3. Put a consumer 24 GB card in the estimating laptop's dock or a second box** and keep the certified card for SOLIDWORKS — two cards, two jobs, no contention with Drawing Search.

What moved on the local model since August

SECONDARY

Qwen3.8-27B

~17 GB at 4-bit

Perplexity ships it locally

Two things. The official model card's agentic numbers are out (Terminal-Bench 2.1 from 63.4 to 73.0, OSWorld-Verified from 63.9 to 84.3), which matters less to us than the vision reports: it reads charts inside PDFs, and when asked for bounding boxes it wrote its own program and returned pixel coordinates, mostly correct. That is the drawing-page task in miniature.

And **Perplexity now ships a post-trained Qwen3.8-27B to run on a customer's Mac** as the private half of a cloud task, with a 0.6B PII classifier deciding what may leave the machine. A search company betting its privacy story on this exact model, at this exact size, is stronger evidence for the class than any benchmark table.

→ THE TRIAL PLAN FROM AUGUST STANDS; ONLY THE CARD QUESTION IS NEW

What the trial would measure, unchanged

The same pages of one pack through `llm_full_extract` on `grok-4.3` and through a local Qwen3.8, against the manual estimate. The pack is now **7332-01 rather than 10575-02**, because 7332-01 has a saved run, a returned manual sheet and a costed record to diff against line by line.

The engine treats vision as rank 40 — it fills gaps and never overrides a DXF or a title-block read — so a worse local model cannot corrupt an estimate; it can only leave more gaps. That is what makes the trial safe, and it is why the answer is a count of gaps filled correctly, not a benchmark score.

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FRONTIER MODELS — FOUR IN 72 HOURS, AND WHAT EACH IS FOR HERE

The engine's rule stands: deterministic readers win, models fill gaps

Claude Fable 5.1 — cheaper to remember, stricter to call

1 SEP 2026

\$10 / \$50 per 1M

Cache read \$0.25

1M context

Watermarked output

The headline is the cache: a cached input token drops from \$1.00 to \$0.25, which turns a long-running agent that re-reads the same context into a quarter of the bill. The three breaking changes are the part a developer has to read: **forced tool choice returns HTTP 400**; earlier models cannot read 5.1's thinking blocks, so a Sonnet-to-Fable escalation chain breaks; and editing an earlier turn invalidates later thinking. Outputs carry an invisible statistical watermark readable through a detection API.

For the engine: `_call_anthropic_llm` sends one plain user message with no tool choice, so none of the three changes apply. The August recommendation stands — the right model on that path is `claude-opus-5` at half the price with the effort toggle, not Fable.

→ NO CHANGE; THE OPUS 5 SWAP IS THE ACTION, AND IT IS CARRIED FROM AUGUST

Gemini 3.8 Flash — the one to try on a whole PDF

2 SEP 2026

\$0.75 / \$3.75 intro

Batch / Flex half

PDF input

1M in · 64K out

Same price as 3.7 Flash to 31 December, then doubling. What is new for us is **native PDF input** with a 1M context: a whole drawing pack can go in as a document rather than page by page as rasters, with a thinking level set per call. Batch and Flex pricing at \$0.375/\$1.875 is the cheapest capable vision on the market for anything that can wait a few minutes.

That is the shape of the **multi-PDF pack** work already in the engine (working, needs testing, "against our model or purely against an LLM"). This is the purely-against-an-LLM arm of that test, and it now has a price a cost case can carry — with a January line for the doubling.

→ WHEN THE MULTI-PDF PACK IS TESTED, 3.8 FLASH ON BATCH IS THE HOSTED COMPARATOR

GPT-6 Astra — frontier price, no reason to move

3 SEP 2026

\$10 / \$50 per 1M

1.05M context

Image input

Effort low → max

OpenAI's first GPT-6, delayed after the July incident to add safeguards, shipped as a partner preview on 3 September and to paid users a day later with cyber capabilities gated. Same sticker as Fable 5.1, cached input \$1, Fast mode at double. Available through the OpenAI API and AWS at identical rates.

The engine's OpenAI path runs `gpt-4.1-mini` as a fallback for price lookups. A frontier-priced model on a fallback path is the wrong trade, and nothing GPT-6 does for drawing pages is claimed to beat the vision routes already on the shortlist. Recorded so nobody has to ask.

Grok 4.6 pricing, now published — and the swap rule

SECONDARY

\$2 / \$6 per 1M

\$4 / \$12 above 200K

500K context

Short-context rates are \$2.00 input, \$0.50 cached, \$6.00 output; a prompt over 200K tokens doubles all three. The engine's vision prompts are single pages, nowhere near the step, so the cheap tier is the real one.

The rule for the two carried swaps, now that a parallel run is live: the five jobs a week are being compared to estimators' sheets on [grok-4.3](#). Changing the vision model under a live batch would put a model change and a review finding in the same diff. So: make the swaps on a **replay of 7332-01 from its saved JSON**, confirm item counts and the costed record hold across three runs, then move the live run to the new strings between batches, one swap at a time.

→ **REPLAY FIRST, ONE SWAP PER RUN, LIVE BATCH UNTOUCHED UNTIL IT IS MEASURED**

I

CNC AUTOMATION FROM DXF — WHAT "AI G-CODE" MEANS IN 2026, AND WHAT IT DOES NOT

For the manufacturing workstream Muhammad is writing up from this month

Every AI CAM product on the market is a milling product

CloudNC CAM Assist — reinforcement learning over machining scenarios, in use at over 1,000 shops — plugs into Mastercam, Fusion, GibbsCAM, SolidCAM and Siemens NX and generates *3-axis and 3+2 milling* strategies. **Toolpath** (Bedford, Ohio; founded 2021) quotes, checks manufacturability and programs in the browser, then hands the CAM program to Fusion — milling. **Autodesk Assistant** in Fusion now creates setups and generates toolpaths from natural language — milling first. None of them lists a 2D laser, punch or plasma post as the thing they automate.

For a TRUMPF laser the automation already exists and is not AI. TruTops Boost imports DXF, STEP and IGES, disassembles assemblies, and its Boost button runs nesting, tooling and sequencing to an NC program with no model in the loop — a 2026 industry assessment says so plainly: rule-based, no AI inside. Oseon sits above it with open interfaces to ERP. SDI runs this stack on SDI-APP01 today.

So the workstream's first question is not "which model writes G-code". It is: what does Boost need to receive, from where, to run its own automation without a person re-keying it — a DXF flat with bend lines on the right layers, a quantity, a material, a due date, and a route. That is exactly the record Muhammad's mapping of the live production spreadsheet produces, and exactly what C1's DXF export and the estimating engine's BOM already carry.

TOOL	WHAT IT DOES	CUT TYPE	READ FOR US
TruTops Boost + Oseon VENDOR	DXF/STEP/IGES in; automatic nesting, tooling, sequencing to NC at one button; Lean Nest processor; shared order pool. Oseon: production control with open ERP interfaces.	Laser · punch · bend	ALREADY OWNED · FEED IT
SPI TRUMPF Interface VENDOR	SPI's SOLIDWORKS sheet-metal add-in with a direct interface to TruTops — unfolds and hands the flat to TRUMPF with bend data intact.	Laser · bend	PRICE AGAINST C1
SigmaNEST · Lantek Expert Cut VENDOR	True-shape nesting and NC programming for laser, plasma, waterjet, punch. Rule-based; Lantek's "Smart Nesting" is engineer-tuned, not learned.	Laser · plasma · punch	COMPARATOR ONLY

TOOL	WHAT IT DOES	CUT TYPE	READ FOR US
CloudNC CAM Assist VENDOR	AI strategy and toolpath generation inside Mastercam / Fusion / GibbsCAM / SolidCAM / NX. 3-axis and 3+2. 1,000+ shops.	Milling	NOT OUR CUT
Toolpath SECONDARY	Browser AI for quoting, DFM and CAM programming; sends the finished program into Fusion.	Milling	NOT OUR CUT
Autodesk Fusion fabrication VENDOR	Setups, tabs, DXF, toolpaths and NC for routers, waterjet, plasma and laser from a post-processor library; Autodesk Assistant drives CAM by natural language.	Laser · plasma · router	IF EVER NOT TRUMPF
ivacam · dxf2gcode SECONDARY	Open source (ivacam: Rust, GPL-3; GRBL/LinuxCNC) DXF/SVG to G-code for mills, lasers, plasma, drag knives, with simulation.	Hobby-class laser	NOT FOR A TRULASER

Where the engine already touches this

The estimating engine reads DXF geometry today for cut length, pierce count and bend count — the same geometry a laser post consumes. It does not need to become a CAM system; it needs the DXF it reads to be the DXF Boost receives, so the estimate and the NC program describe one part. C1's flat-pattern export with bend lines is that file. The production report and the actuals coming back are the other half of the loop this workstream closes.

→ INTO MUHAMMAD'S BRD AS QUESTIONS: WHAT BOOST IMPORTS, ON WHICH LAYERS, AND THE OSEON ORDER INTERFACE

What to stop saying

The programme pages describe this workstream as *"AI-generated G-code from DXF geometry, streamed to CNC machines"*. In 2026 that sentence is true of milling shops and not of a TRUMPF laser line, where the programming is already automatic and the gap is the order data around it. The write-up should describe the integration — DXF, quantity, material, route into Boost's order pool; actuals back out of Oseon — and leave G-code to the post-processor TRUMPF already ships. That is a smaller claim and a deliverable one.



DRAWING PACKS FROM SOLIDWORKS — LEO DRAWS BY CONVERSATION FROM OCTOBER

C3 is POC-complete in-house; the vendor is about to ship the interactive version

SOLIDWORKS 2027 lands in October with drawing generation in the box

The Preview programme ran from 15 July to mid-August; the launch event is **SOLIDWORKS Live on 15 October**. In 2027, **LEO generates a drawing from a model**: you state the template, the standard, the primary view and what the drawing must carry, refine it by prompt, and save to a location. The 3DEXPERIENCE beta it grew from produces standard and section views, hole callouts, driven dimensions, GD&T frames and datums, revision tables and BOMs for assemblies. Autodesk has the same in Fusion as Drawing Automation.

This overlaps C3 directly (drawing generation from ASM/PRT, POC-complete, waiting on seats). It does not replace it: LEO is a conversation per model; P2 needs a batch across a project folder with no one typing. The decision is which of the two SDI owns for the batch case, and whether the answer is "LEO for the designer at the desk, C3 headless for the production pack". Also unresolved: whether LEO runs on the desktop seat or needs a 3DEXPERIENCE role — the Labs tab in 2026 FD01 is desktop, so probably the former, but confirm.

LEO's other trick is still F1, and still untried

VENDOR

August's first action was to open SOLIDWORKS Labs on a seat we hold and try LEO's PDF-to-parametric on a legacy drawing, because F1 on the roadmap is that exact capability marked "POC pending". No result is recorded. Since then Dassault added STEP and IGES to parametric in 2026x FD02, so the reverse-engineering path now covers both ways a pack arrives without native models.

A permanent seat is now installed on the estimating laptop. The trial costs an hour.

→ **CARRIED: ONE LEGACY PDF THROUGH LEO, ON THE LAPTOP SEAT, RESULT WRITTEN DOWN**

The Document Manager API key moves with the release

VENDOR

P1 and P2 are built on the Document Manager API, and a new API key must be requested for each major release. When designers move to SOLIDWORKS 2027, files saved from it will be read by a 2026 DM library until that key and library are updated. This is a diary item, not a risk — but it lands in the same window as P2's go-live on 29 September and the PDM MVP build, and it is cheaper to schedule than to discover.

→ **ON YOGESH'S LIST: REQUEST THE 2027 DM API KEY AT RELEASE; TEST P1/P2 AGAINST 2027 FILES**

The add-in route to the flat pattern, one more entry

VENDOR

August named CADshift for batch DXF export with bend lines. **SPI's TRUMPF Interface** is the other one worth a price: an SDI-relevant vendor whose sheet-metal add-in unfolds in SOLIDWORKS and hands the flat directly to TruTops with bend data intact — which joins the C1 question to the CNC one above. Both need a seat, as C1 does; the question remains whether to own that export or buy it.

→ **PRICE SPI ALONGSIDE CADSHIFT; 20 MINUTES WITH YOGESH, CARRIED FROM AUGUST**

I

ESTIMATING — THE MARKET GOT CLOSER, AND ONE PRODUCT IS WORTH AN AFTERNOON

Drawing-to-quote is now a category; nobody in it prices against SDI's own sheet

Werk24 — a PDF drawing in, structured JSON out

VENDOR

€0.95 / page, no commitment

or €990 / month + €0.85 / page

5–20 s / page

Python client

Reads a PDF or image drawing and returns JSON in three mixable SKUs: the **title block** (drawing ID, designation, material, weight, tolerances, projection, BOM, notes); **features** (dimensions, fits, threads, bores, chamfers, GD&T, radii, surface roughness, each with a confidence and x/y bounding box); and **process signals** (primary and secondary processes to DIN 8580, volume and bounding-box estimates, feasibility flags for quoting). CAD approximation currently for flat parts, which is to say sheet metal.

This is the first product that does what `llm_full_extract` and `extractor_patterns` do, rather than what a construction takeoff tool does. It does not price, does not know the SDI workbook, and does not resolve a GA-to-detail tree. What it offers is an independent read of the same page with a confidence per field — either a rank-40 source beside vision, or the comparator the engine has never had.

→ AN AFTERNOON: THE 7332-01 PACK THROUGH THEIR API; THEIR JSON AGAINST THE COSTED RECORD, FIELD BY FIELD

Paperless Parts Wingman, and the rest of the category

SECONDARY

Wingman

QuotationFactory

CutQuote

aPriori

Wingman reads the print — bends, welds, threads, finishes, critical callouts — and flags risk before the estimator prices; a proprietary model, from around \$500 a month per shop.

QuotationFactory (Netherlands) prices laser-cut sheet parts from geometry with production parameters. **CutQuote** turns a DXF into a laser quote in seconds — cut length and pierces, nothing after the laser. **aPriori** costs from a digital twin of the part for large-manufacturing buyers.

The verdict from August holds and is now sharper: the category has moved from measuring floor plans to reading fabrication prints, so the *approach* is validated by a market; and still nobody prices a folded, welded, plated display unit on SDI's workbook against 1,982 SDI jobs with a source on every line.

The research route, if the readers plateau

SECONDARY

Two published systems are the fine-tune path: **Florence-2 fine-tuned on 400 annotated drawings** for GD&T, benchmarked against GPT-4o and Claude 3.5 on precision, recall and hallucination; and a **hybrid of YOLOv11-obb oriented detection with a Donut document parser** over 1,367 drawings across nine field classes (GD&T, general tolerances, measures, materials, notes, radii, roughness, threads, title blocks). Oriented boxes matter because dimensions sit at angles and title blocks stack.

Not an action. SDI's own corpus of drawings with returned manual estimates is exactly the training set these papers had to build by hand — a reason to keep the parallel run's returned sheets, not just the verdicts.

Document models, one line each

SECONDARY

Mistral OCR 4 returns text with bounding boxes, typed blocks (tables, titles, signatures) and inline confidences. **dots.ocr** parses multilingual layout in a single vision-language model. The consensus on classical tools has not moved: pdfplumber, Camelot and Docling are right for digital PDFs with borders; layout models are for scans and borderless tables. The engine's hybrid — rules for the predictable fields, a model for the rest — is still the recommended shape.

I

REPLICATING PDM IN-HOUSE — THREE THINGS TO DESIGN IN BEFORE P3.1

MVP now 6 November after the 4 September re-plan; the build case is unchanged

GitLab already gives you check-out enforcement for free

VENDOR

Git LFS exclusive locks

.gitattributes lockable

SOLIDWORKS honours read-only

Mark `*.sldprt` and `*.sldasm` as `lockable` in `.gitattributes` and Git LFS makes every local copy **read-only until locked**; GitLab exclusive locks hold across all branches. SOLIDWORKS warns on opening a read-only file, which is the closest a Git workflow gets to PDM-style check-out — and it is the exact behaviour P3.1's "check-in/out and locking" describes.

This is the cheapest decision on the PDM: adopt LFS locks as the mechanism in the MVP rather than writing a lock table, and spend the effort on the part nothing sells — the Document Manager intelligence layer that knows what is inside the file.

→ **P3.1 DESIGN NOTE: LFS EXCLUSIVE LOCKS ARE THE LOCK; THE INTELLIGENCE LAYER SITS ABOVE THEM**

Anchorpoint, read as prior art

VENDOR

Locks a thousand files in under a second through its own metadata server, which **can be self-hosted**; cleans Git LFS caches to keep storage down; sparse checkout for TB-scale repositories; SOLIDWORKS handled without configuration; monthly releases. August asked for an hour on how it handles binary churn; no result is recorded. The LFS-cache clean-up is their answer to the storage half of it — prune unreferenced objects on a schedule — and it is the one to copy.

→ **CARRIED: AN HOUR ON ANCHORPOINT'S CHURN AND LOCKING MODEL BEFORE P3.1 STARTS**

Aras Innovator Community Edition — free, and not the answer

VENDOR

No licence fee up to 50 named users. But the SOLIDWORKS connector is Elysium's, sold separately; the 3D visualisation release supports SOLIDWORKS to 2025; and it is enterprise PLM with a configuration surface nobody at SDI would own. Worth reading for its data model — parts, documents, CAD structure, where-used — before P3.2 designs ours. Not worth installing.

The binary-churn question, restated for November

August's warning stands: a SOLIDWORKS assembly changes at the binary level on every open-and-save, so a naive LFS history grows with activity rather than with change. With P3.1 moved to 6 November there is more time to answer it, not less reason: gate commits on explicit check-in rather than on save, content-hash the meaningful payload, and prune on a schedule. Anchorpoint does the third; GitLab locks make the first natural.

WHAT TO ACTUALLY DO

In order. Item 1 is this week's; the rest are a day or less each.

#	ACTION	EFFORT	WHY NOW
1	Decide the workstation's card before the order goes	A conversation	The 16 GB certified card in the spec cannot run the 27B local vision trial. Keep vision hosted, step to a 24 GB certified card, or put a consumer card elsewhere — any of the three is fine, but it should be chosen and not discovered.

#	ACTION	EFFORT	WHY NOW
2	Make the Opus 5 and Grok 4.6 swaps on a replay of 7332-01	Half a day	Carried from August, still in the code as <code>claude-sonnet-4</code> and <code>grok-4.3</code> . Replay from the saved JSON, one swap per run, three runs each; the live parallel run keeps its strings until measured.
3	Run the 7332-01 pack through Werk24's API	An afternoon · under £20	First product that reads a fabrication drawing into typed JSON with confidences. Their fields against the costed record tell us where our readers are weak, per field, for the price of a few pages.
4	Put TruTops Boost's import and Oseon's order interface into Muhammad's BRD as questions	An hour	The CNC workstream's deliverable is the order record into Boost and the actuals back, not G-code. Naming that now keeps the BRD pointed at something that can ship on the stack SDI owns.
5	Try LEO on one legacy PDF, on the laptop seat	An hour	Carried from August with no result recorded. F1 is still the most expensive roadmap item and this is still its cheapest test; 2027's drawing generation makes the same seat worth opening twice.
6	Diary the SOLIDWORKS 2027 Document Manager API key	Minutes	A new key per major release; P1 and P2 read files through it. Release is October; P2 goes live 29 September. Schedule the key request and a

#	ACTION	EFFORT	WHY NOW
			test against 2027-saved files.
7	Write "LFS exclusive locks are the lock" into the P3.1 design	An hour	Read-only-until-locked is already what GitLab and SOLIDWORKS do together. Not building a lock table is a week saved on an MVP with one developer.
8	Price SPI's TRUMPF Interface alongside CADshift	20 minutes	Two add-ins now cover C1's flat-pattern export, one of them straight into TruTops. Buy-versus-own for C1 is Yogesh's call and it is cheaper with two prices on the table.
9	Keep the Gemini January line	Minutes	3.8 Flash inherits 3.7's introductory rate to 31 December and doubles on 1 January 2027. Any multi-PDF cost case needs that line.

What is deliberately not in this report

Nothing here proposes a model where a deterministic reader works, nothing proposes the engine become a CAM system, and nothing proposes buying a PDM. Four frontier models shipped in three days and the recommendation for the engine is two model-string swaps that were already due, made on a replay so the live parallel run is not disturbed. **The interesting technology this month sits at rank 40 or beside the engine, not inside its arbitration — and that is where it should stay.**

SDI Intelligence — AI & Technology Landscape, September 2026. Issued 7 September 2026 · James Gray.
Supersedes the August issue of 26 August 2026.

Compiled from web research on 7 September 2026, read against the engine's own code ([config.py](#) , [web_ai_price_lookup.py](#) , [llm_scan_price.py](#)) and the shared workstation specification of 3 September.
Vendor-sourced entries: anthropic.com and platform.claude.com (Fable 5.1 migration guide), solidworks.com and

blogs.solidworks.com (2027 preview, LEO), trumpf.com (TruTops Boost, Oseon), cloudnc.com, autodesk.com, werk24.io (pricing, API), paperlessparts.com, spi.de, sigmanest.com, lantek.com, docs.gitlab.com (file locking), anchorpoint.app, aras.com, mistral.ai, huggingface.co (Qwen, Perplexity). Secondary sources include venturebeat.com, marktechpost.com, codersera.com, eesel.ai, benchlm.ai, mem0.ai, theregister.com, 9to5mac.com, hawkridgesys.com, goengineer.com, vanenkhuisen.com, engineering.com, demystifyingplm.com, mindstudio.ai, yottalabs.ai, en.wikipedia.org (GPT-6 Astra) and arxiv.org (2411.03707, 2505.01530, 2506.17374).

Figures marked Secondary are third-party reports and are not vendor-confirmed. Model pricing and availability move quickly; confirm against the vendor's own pricing page before committing spend or hardware. Introductory rates carry end dates. The 17 GB figure for the 4-bit 27B model is a secondary report and should be checked against the model card before a card is chosen on it.